

PrivateFin: A Privacy-First Explainable Financial Advisory System

Background Research & Literature Review — Topic 3

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1. Introduction

The application of Large Language Models (LLMs) to retail finance has exposed a Transparency-Privacy Gap: high-capability AI advisory systems are simultaneously opaque (lacking auditable reasoning traces), privacy-invasive (requiring cloud inference), and inaccessible (proprietary deployment). PrivateFin is designed to close this gap by combining a quantised open-source LLM (Meta's Llama 3, served locally via Ollama) with a deterministic market-data pipeline (yfinance, Pandas) and Structured Chain-of-Thought (CoT) prompting. This survey evaluates the academic and technical foundations of this design across four domains: domain-specific LLM architectures (2), specialised NLP methodology (3), quantitative frameworks (4), and the regulatory landscape that transforms explainability from a desirable property into a legal obligation (5). Each section identifies the specific gap that PrivateFin addresses; 6 synthesises these into a unified research contribution.

2. Domain-Specific Large Language Models

2.1 BloombergGPT (Wu et al., 2023)

Architecture and Training

BloombergGPT is a 50-billion parameter, decoder-only causal language model trained on FinPile—a 363-billion token hybrid corpus comprising 51% proprietary Bloomberg data (earnings transcripts, analyst reports, regulatory filings) and 49% general internet corpora (The Pile, C4, Wikipedia). Its distinguishing contribution is this domain-weighted corpus construction rather than architectural novelty. The model is trained via a standard Next Token Prediction objective, but the proprietary slice is chronologically ordered, exposing the model to temporally coherent financial event sequences that shuffled internet corpora cannot replicate. Distributed training employed tensor and pipeline parallelism across GPU clusters. On five domain-specific benchmarks—including Financial PhraseBank (FPB) sentiment, FiQA question answering, and financial named entity recognition—BloombergGPT consistently outperforms GPT-NeoX-20B and OPT-66B despite comparable or smaller parameter counts, validating domain-weighted training as a more efficient approach than raw scale for financial tasks.

Critical Evaluation

BloombergGPT's limitations are structural. It is deployed exclusively as a proprietary SaaS product accessible via Bloomberg Terminal subscriptions (>£20,000 p.a.), creating an access gatekeeping problem. More critically for PrivateFin's design, cloud-side inference means every query transmits user intent—which securities are researched, at what frequency, and in what context—to external servers. Under GDPR Article 4 (1), such query data constitutes personal data where it is reasonably linkable to a natural person (GDPR, 2016), and Bloomberg's terms do not preclude its use for commercial model improvement. Finally, BloombergGPT produces no structured reasoning output: recommendations are generated auto regressively without enforced logical scaffolding, constituting the 'black-box' behaviour the EU AI Act (2024) designates non-compliant for high-risk AI applications. PrivateFin addresses all three failures: local execution eliminates data transmission; CoT prompting mandates reasoning traces.

2.2 FinGPT (Yang et al., 2023)

Architecture and Design

FinGPT adopts a parameter-efficient fine-tuning paradigm built on Low-Rank Adaptation (LoRA). LoRA decomposes weight update matrices into two low-rank matrices ($W = W_0 + BA$, where $B \in \mathbb{R}^{(d \times r)}$ and $A \in \mathbb{R}^{(r \times k)}$, with $\text{rank } r \ll \min(d, k)$), reducing trainable parameters by over 99% relative to full fine-tuning. This makes domain adaptation feasible without Bloomberg-scale compute budgets. The pipeline has three stages: (1) real-time ingestion from SEC EDGAR, Twitter/X, Reddit, and news APIs; (2) LoRA fine-tuning of foundation models (LLaMA, Falcon, ChatGLM) on financial instruction-following and sentiment objectives; and (3) evaluation on FPB, FiQA-SA, and TFNS benchmarks. LoRA-fine-tuned LLaMA-7B variants achieve competitive performance against much larger general-purpose models, validating targeted adaptation over raw scale—a principle directly adopted by PrivateFin.

Critical Evaluation

FinGPT's open-source philosophy is its stated strength, but its practical deployment reconstructs the very cloud dependency it claims to avoid: documented implementations route inference through OpenAI GPT-3.5/4 APIs

or Hugging Face Inference endpoints. Data sovereignty is therefore not achieved. Furthermore, FinGPT ingests speculative social media discourse (Reddit, Twitter) alongside structured SEC filings without provenance weighting, making its recommendations un-anchored to verifiable market data. When a user asks 'why this stock?', FinGPT cannot produce a metric-traceable answer. PrivateFin resolves this by grounding all LLM reasoning in outputs from a deterministic quantitative layer, ensuring every justification references auditable, reproducible calculations.

3. Specialised NLP: FinBERT (Araci, 2019)

3.1 Architecture

FinBERT applies domain-adaptive pre-training (DAPT) to BERT (Devlin et al., 2019), an encoder-only transformer architecture. Unlike the autoregressive decoders above, BERT's bidirectional self-attention conditions each token's representation on all surrounding tokens simultaneously, producing contextualised embeddings well-suited to discriminative classification. Araci (2019) initialises from BERT-base-uncased weights and continues pre-training on a financial corpus comprising Reuters TRC2 articles (1.8M documents), earnings transcripts, and analyst reports—adapting internal representations to financial vocabulary (EBITDA, basis points, P/E ratios) and the hedged, forward-looking language that general sentiment lexicons misclassify. A soft max classification head on the [CLS] token is then fine-tuned on the Financial PhraseBank (Malo et al., 2014), yielding 97.1% accuracy and 96.8% F1—markedly above VADER and vanilla BERT, demonstrating that domain-specific pre-training yields disproportionate gains even at small model sizes.

3.2 Relevance and Limitations

FinBERT establishes the foundational principle that domain-targeted training outperforms general-purpose models even with fewer parameters—a principle PrivateFin extends to prompt engineering: saturating the LLM's context with structured financial indicator data produces more accurate qualitative interpretations than vague natural language descriptions. However, FinBERT is architecturally incapable of generative tasks: it outputs a probability distribution over three sentiment classes, which does not constitute an explanation in any regulatory sense. Its typical inference pathway also relies on Hugging Face cloud checkpoints. PrivateFin synthesises FinBERT's domain-specificity insight with the generative and conversational capability of a decoder-only LLM, while retaining full local execution.

4. Quantitative Data Ingestion and Technical Indicators

4.1 yfinance: Privacy-Preserving Market Data

The yfinance library (Aroussi, 2019) retrieves OHLCV (Open, High, Low, Close, Volume) time-series data from Yahoo Finance as Pandas DataFrames, parameterised by ticker, date range, and interval. Critically for a privacy-preserving system, yfinance requires no API key registration, eliminating the creation of user-linked data trails on third-party servers—a property absent in commercial alternatives (Alpha Vantage, Bloomberg API, Quandl). Its determinism is equally important: given identical parameters, yfinance returns identical data, enabling any historical recommendation to be independently audited. This reproducibility is foundational to PrivateFin's compliance with the EU AI Act's traceability mandate: the quantitative inputs to every LLM prompt can be independently verified from public market data.

4.2 Technical Indicator Computation (Pandas)

Two indicators, computed deterministically via Pandas (McKinney, 2010), serve as the primary structured inputs to PrivateFin's reasoning layer. The Relative Strength Index (RSI; Wilder, 1978) is defined as $RSI = 100 - [100 / (1 + RS)]$, where RS is the ratio of the average gain to average loss over a rolling n-period window (standard $n = 14$). $RSI \in [0, 30]$ signals oversold conditions; $RSI \in [70, 100]$ signals overbought conditions. The MACD (Appel, 1979) is computed as $EMA(12) - EMA(26)$, with a signal line $EMA(9)$ of the MACD series; a bullish crossover occurs when MACD crosses above the signal line. Both indicators are closed-form and deterministic: given the same OHLCV input, outputs are always identical. In PrivateFin, these values anchor the CoT prompt—providing objective, verifiable data points that the LLM must reference in its reasoning chain—and constitute the audit trail satisfying the EU AI Act's Article 13 transparency requirement.

5. Regulatory and Ethical Landscape

5.1 EU AI Act (2024): Explainability as Legal Obligation

The EU AI Act (Regulation (EU) 2024/1689) classifies AI systems used for investment advisory as High-Risk AI Systems (HRAIS) under Article 6 and Annex III. This triggers obligations under Chapter III: Article 13 requires HRAIS to be sufficiently transparent that users can interpret outputs and use them appropriately; Article

17 mandates technical documentation enabling reconstruction of the system's reasoning; Article 14 requires human oversight mechanisms. Critically, the Act's recitals specify that explainability must be meaningful—a disclosure stating 'generated by a neural network' is insufficient. The system must articulate the specific features and reasoning steps informing each output. PrivateFin's Structured CoT prompting satisfies this standard: each prompt instructs the model to enumerate relevant indicators, their current values, the thresholds evaluated against them, and the logical inference connecting data to recommendation.

5.2 GDPR: Data Sovereignty Architecture

Three GDPR provisions directly shape PrivateFin's architecture. Article 22 grants data subjects the right not to be subject to decisions based solely on automated processing with significant effects—investment recommendations clearly qualify. Compliance requires that the system be capable of producing an explanation of the decision, which PrivateFin's CoT output satisfies. Articles 5 and 25 establish data minimisation and Privacy by Design respectively: Article 25 mandates that data protection principles be embedded in system design, not applied post hoc. Cloud inference is structurally incompatible with this: it requires transmitting financial query data to external processors. Research by the FCA (2023) found that 68% of retail investors are unaware their interaction data is used for model training, making informed consent unreliable as a compliance mechanism. PrivateFin's Ollama-based local inference satisfies Articles 5 and 25 by architectural construction—no personal data leaves the user's device.

6. Synthesis and Research Gap

The literature reveals a bifurcated landscape: systems are either high-performance but proprietary and cloud-dependent (BloombergGPT), or open-source but still cloud-reliant at inference time with no deterministic auditability (FinGPT). FinBERT demonstrates the value of domain adaptation but is generatively incapable and similarly cloud-deployed. No existing system integrates local LLM inference, deterministic quantitative grounding, and structured CoT explainability within a single privacy-preserving architecture.

PrivateFin occupies this gap through four integrated components: (1) quantised Llama 3 via Ollama, enabling LLM-class reasoning on consumer M1 hardware without cloud transmission; (2) a yfinance + Pandas pipeline generating reproducible, auditable technical indicators; (3) Structured CoT prompting producing metric-anchored reasoning traces that satisfy EU AI Act Article 13; and (4) full Privacy by Design compliance under GDPR Articles 5 and 25. The combination of these components constitutes a demonstrably novel contribution: the first retail-facing, fully local, explainable financial advisory system grounded in deterministic market data.

7. References

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